

TECHNOLOGIA 2.0

Round 2 · Operation Phoenix

MISSION 1 OF 4

Project Discovery

Suggested time: ~15 minutes | Deliverable: Technology selection & requirement analysis

You are the Graduate Software Engineer assigned to **Project Phoenix**: rebuilding the technology platform for **NimbusCart**, a growing online marketplace connecting local sellers with regional buyers.

Client Requirements

- Sellers can register, create a storefront, and list products with images, prices, and descriptions.
- Buyers can browse, search, and filter products, then complete a purchase through checkout.
- Every user (seller or buyer) logs in securely to a single account.
- Buyers and sellers receive notifications for key order events (order placed, shipped, delivered).
- Checkout must support real payment processing, not just a placeholder form.
- The operations team needs basic visibility into whether the platform is healthy at any given time.

Business Constraints

- **Budget cap:** 9,500 budget units per month total (≈ ■8,00,000/month), across every technology tier combined — this cap applies for the rest of the simulation too.
- **Team size:** a small engineering team with limited dedicated DevOps capacity — prefer managed services over heavy self-managed infrastructure unless there's a strong reason not to.
- **Launch scope:** initial launch is single-region, with moderate expected traffic — this is a pilot, not day-one hyperscale.
- **Compliance:** any component touching payment card data must be PCI-DSS compliant.
- **Timeline:** the client wants a working platform soon; favor proven, fast-to-integrate options over long build-outs.

Technology Inventory

Review the available options below. Unfamiliar terms are covered in the Glossary & Hints booklet.

Category	Option	Cost	Engineering Note
Frontend App Tier REQUIRED	React / Next.js SPA (CDN Hosted)	150	Fast static delivery via global edge CDN with serverless SSR support.
	Server-Rendered Templates (Monolithic UI)	300	Couples UI closely to backend app instances; harder to scale independently.

Category	Option	Cost	Engineering Note
Backend API Engine REQUIRED	Auto-Scaling Managed Compute (VM / Containers)	1500	Managed environment with low ops overhead for application services.
	Managed Kubernetes Cluster	3000	High operational complexity and cost; steep learning curve for small teams.
	Serverless Functions (FaaS)	600	Cost-effective for sporadic traffic, but vulnerable to cold starts.
Primary Database REQUIRED	Managed Relational DB — Medium + Read Replica	900	Separates primary read/write traffic while maintaining transaction integrity.
	Managed Relational DB — Multi-AZ + 2 Replicas	2600	High availability setup, best suited for heavy production loads.
	Managed NoSQL Document Store	800	Great for flexible schemas; requires extra care for relational data integrity.
Identity & Authentication REQUIRED	Managed Auth Service (OAuth2 / OIDC)	120	Offloads session security, MFA, and credential storage compliance.
	Custom JWT Auth Service	50	Requires full team maintenance for security patches and token rotation.
Notifications & Message Queues REQUIRED	Managed SQS / Message Broker	100	Asynchronously decouples background tasks (emails, webhooks, processing).
	Distributed Event Stream (Kafka-class)	1400	High throughput event stream; high maintenance overhead.
Observability & Analytics optional	Managed Log Aggregation & Metrics Suite	250	Provides real-time dashboards, alerting, and APM tracing.
	Basic Cloud Provider Metrics	50	Includes fundamental CPU/Memory metrics without deep application tracing.
Payment Gateway Integration optional	PCI-DSS Certified Payment API	0	Per-transaction fee structure — zero infrastructure management overhead.
	In-House Vault Storage	500	Violates compliance guidelines unless rigorous audit standards are met.

The AI Recommendation Engine, Caching, Storage & CDN, and Disaster Recovery tiers are not required for this mission — they appear later in the Technology Inventory as the project evolves.

Deliverable

- Select one option for each **required** category above.
- For each choice, write a brief justification (1–2 sentences) tying it back to the client requirements or business constraints.
- Write a short requirement analysis summary (2–4 sentences) explaining how your selections meet the client's needs within the budget cap.

Evaluation Focus

Requirement Analysis • Technology Selection